

MA1521 L1 Week 04 Wed 20260902

Find the point(s) on each of the following surfaces which is closest to the origin.

a) $xyz^2 = 1$

b) $x^2 - yz = 1$

a) Surface is the set of points
 (x, y, z) satisfying $xyz^2 = 1$

Want to minimize distance f_n

$\sqrt{x^2 + y^2 + z^2}$ for pt on surface

Equiv: minimize the f_n $x^2 + y^2 + z^2$

i.e. f_n $D(x, y) := x^2 + y^2 + \frac{1}{xy}$

Here $D_x = 2x - \frac{1}{x^2 y}$

$D_y = 2y - \frac{1}{xy^2}$

At critical pt (x_0, y_0) (where min D occurs)

$D_x(x_0, y_0) = 0$ i.e. $2x_0 - \frac{1}{x_0^2 y_0} = 0$

$D_y(x_0, y_0) = 0$
 $2y_0 - \frac{1}{x_0 y_0^2} = 0$

get $x_0^2 = y_0^2$ from $(*)$
 so $y_0 = +x_0$ or $-x_0$

If $y_0 = x_0$, then $(*)$ gives
 $x_0^4 = \frac{1}{2} = y_0^4$

$$\leftarrow \begin{aligned} &\text{e. } 2x_0^3 y_0 = 1 = 2x_0 y_0^3 \quad \text{--- } (*) \\ &\text{i.e. } x_0^2 = \frac{1}{2} = y_0^2 \quad \text{X should be } x_0^4 = \frac{1}{2} = y_0^4 \\ &\text{i.e. } x_0 = \pm \frac{1}{\sqrt{2}}, \quad y_0 = \pm \frac{1}{\sqrt{2}} \quad x_0 = \pm \sqrt[4]{\frac{1}{2}}, \quad y_0 = \pm \sqrt[4]{\frac{1}{2}} \end{aligned}$$

If $y_0 = -x_0$, then by $(*)$, get $2(-x_0^4) = 1$, contradiction.

Hence must have $x_0 = y_0 = \pm \frac{1}{\sqrt{2}} \quad \text{X} \quad \text{corresp. } z_0 = \pm \frac{1}{\sqrt{2}} \quad \text{X} \quad z_0 = \pm \sqrt[4]{2}$

4 soln points. $(x_0, y_0, z_0) = \pm \frac{1}{\sqrt{2}} (1, 1, 1) \quad \text{X} \quad \pm \frac{1}{\sqrt[4]{2}} (1, 1, \sqrt{2})$
 and $\pm \frac{1}{\sqrt{2}} (1, 1, -1) \quad \text{and } \pm \frac{1}{\sqrt[4]{2}} (1, 1, -\sqrt{2})$

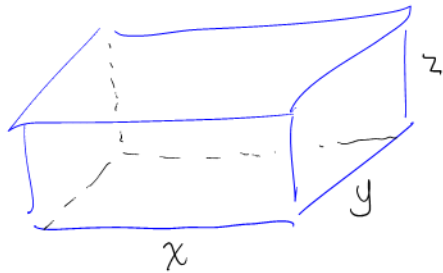
b) $x^2 - yz = 1$

minimize $D(y, z) (= x^2 + y^2 + z^2) = 1 + y^2 + y^2 + z^2$

$$\left. \begin{aligned} D_y &= z + 2y \\ D_z &= y + 2z \end{aligned} \right\} \begin{aligned} &\text{At crit pt. } (y_0, z_0), \text{ get } D_y(y_0, z_0) = 0 = D_z(y_0, z_0) \\ &\text{i.e. } z_0 + 2y_0 = 0 \quad \text{i.e. unique soln } y_0 = z_0 = 0 \\ &\quad y_0 + 2z_0 = 0 \quad \Rightarrow x_0 = \pm 1 \end{aligned}$$

2 soln points $(1, 0, 0)$ and $(-1, 0, 0)$

front/back left/right
 A rectangular produce box is to be made of cardboard; the sides of single thickness, the ends of double thickness, and the bottom of triple thickness, with the top left open. Its volume is to be 1 cubic foot; what should be its proportions in order to use the least cardboard?



Volume is xyz , fixed as 1. ($\Leftrightarrow z = \frac{1}{xy}$)

$A :=$ amt of cardboard $= 2 \times \underbrace{1 \cdot yz}_{\text{sides}} + 2 \times \underbrace{2 \cdot xz}_{\text{ends}} + \underbrace{3 \cdot xy}_{\text{btm}}$
 ↑
 Want to minimize

$$A = \frac{2}{x} + \frac{4}{y} + 3xy$$

Have $\left. \begin{aligned} A_x &= -\frac{2}{x^2} + 3y \\ A_y &= -\frac{4}{y^2} + 3x \end{aligned} \right\} \begin{aligned} &\text{At min pt, have } A_x = 0 = A_y \\ &\text{i.e. } 3y = \frac{2}{x^2} \text{ and } 3x = \frac{4}{y^2} \quad \text{--- } (*) \\ &[[\text{Exploit symmetry!}]]$

i.e. $3xy = \frac{2}{x} = \frac{4}{y}$

$$\Rightarrow y = 2x$$

So at min pt, get (from $*$) $6x^3 = 2$, so $x = \sqrt[3]{\frac{1}{3}}$

$$\text{so } \overset{\text{min pt}}{(x, y, z)} = \frac{1}{\sqrt[3]{3}} \left(1, 2, \frac{3}{2} \right)$$

(Assume known to be min pt)

$$\text{so } y = \frac{2}{\sqrt[3]{3}} \quad \text{and} \quad z = \frac{1}{2\sqrt[3]{3}^2} = \frac{\sqrt[3]{3}^2}{2} = \frac{3}{2\sqrt[3]{3}}$$

$$f(x,y) :=$$

Find the extremal point of $x^2 + 2xy + 4y^2 + 6$ and show it is a minimum point

(1st mthd: find critical pts, use 2nd der. test.)

$$\left. \begin{aligned} f_x(x,y) &= 2x + 2y \\ f_y(x,y) &= 2x + 8y \end{aligned} \right\} \begin{aligned} &\text{At critical points, have } \begin{cases} 2x + 2y = 0 \\ 2x + 8y = 0 \end{cases} \\ &\text{Solve, get } (x,y) = (0,0) \text{ unique soln.} \end{aligned}$$

i.e. f has only 1 crit. pt at $(0,0)$.

Compute $\begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 8 \end{pmatrix}$ $\leftarrow$ always symmetric matrix
bec. $f_{xy} = f_{yx}$

Hessian matrix of f

Recall: If $\det(\text{Hessian}) < 0$, then saddle pt.

$\det(\text{Hessian}) > 0$, and $\begin{pmatrix} \text{top left entry} \\ \text{btm right} \end{pmatrix} > 0$ then min pt

$\det(\text{Hessian}) > 0$, and $\begin{pmatrix} \text{top left entry} \\ \text{btm right} \end{pmatrix} < 0$ then max pt

For Hessian = $\begin{pmatrix} 2 & 2 \\ 2 & 8 \end{pmatrix}$, here $\det = 12$ is > 0 , hence by 2nd der test
 and (top left entry) is > 0 this is min pt
 btm right

$$f(x, y) :=$$

Find the extremal point of $x^2 + 2xy + 4y^2 + 6$ and show it is a minimum point

2nd method: Completing the square!

$$f(x, y) = (x+y)^2 - y^2 + 4y^2 + 6 = (x+y)^2 + 3y^2 + 6$$

$$\left[= (x+y)^2 + 3(y^2 + 2) \right] \neq (x+y)^2 + 3((y+1)^2 - 1)$$

extra $2y$ term left over.

see that

for all $x, y \in \mathbb{R}$, get $f(x, y) \geq 6$

and $f(x, y) = 6$ min val

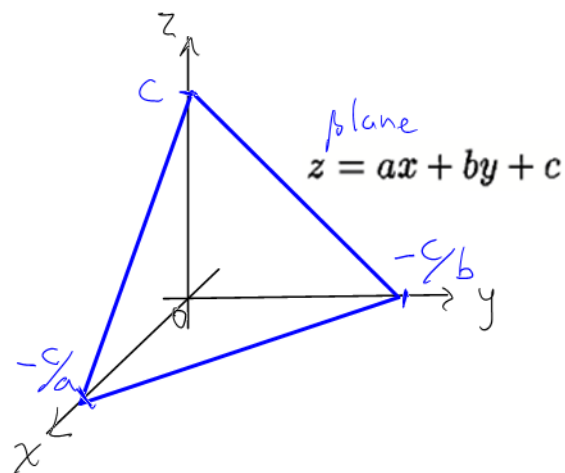
iff $x+y = y = 0$

i.e. iff $(x, y) = (0, 0)$

Consider all planes passing through the point (2,1,1) and such that the intercepts on the three coordinate axes are all positive.

For which of these planes is the product of the three intercepts smallest?

(Hint: take the plane in the form $z = ax + by + c$, where a and b are the independent variables.)
and c



Intercepts $-\frac{c}{a}, -\frac{c}{b}, c$ are all > 0

(so $c > 0, a < 0, b < 0$)

Plane goes thru $(2, 1, 1)$ means $1 = a \cdot 2 + b \cdot 1 + c$
 $\begin{matrix} \uparrow & \uparrow & \uparrow \\ x\text{-val} & y\text{-val} & z\text{-val} \end{matrix}$

$$\text{so } c = 1 - 2a - b$$

Want to find (a, b) such that $\left(\frac{-c}{a}\right)\left(\frac{-c}{b}\right) \cdot c$ is smallest

$$P := \frac{c^3}{ab} = \frac{(1 - 2a - b)^3}{ab}$$

At critical pts of P , know $P_a = \frac{3(1-2a-b)^2 \cdot (-2)ab - (1-2a-b)^3 \cdot b}{a^2b^2}$

$$P_b = \frac{3(1-2a-b)^2 \cdot (-1)ab - (1-2a-b)^3 \cdot a}{a^2b^2}$$

must vanish ($=0$)

Hence, at crit. pt., get $\left. \begin{array}{l} -6a = 1-2a-b \\ -3b = 1-2a-b \end{array} \right\} \Rightarrow b=2a, \text{ so } a = -\frac{1}{2}, b = -1$
and $c=3$

Hence plane is $z = -\frac{1}{2}x - y + 3$ (pdt of intercepts $\bar{P}$ is $6 \cdot 3 \cdot 3 = 54$)

→ Why is this giving smallest $\bar{P}$?

Needs detailed analysis.

HW Exer: redo a), b) using "completing the square".

For each of the following functions, find the critical points, and classify them using the 2nd-derivative criterion.

$f(x,y) =$	a) $x^2 - xy - 2y^2 - 3x - 3y + 1$	b) $3x^2 + xy + y^2 - x - 2y + 4$	c) $2x^4 + y^2 - xy + 1$
$f_x(x,y) =$	$2x - y - 3$	$6x + y - 1$	
$f_y(x,y) =$	$-x - 4y - 3$	$x + 2y - 2$	
At <u>crit. pt.</u> get Solve for (x,y)	$2x - y = 3$ $-x - 4y = 3$ Soln is $(x,y) = (1, -1)$	$6x + y = 1$ $x + 2y = 2$ Soln is $(0, 1)$	
Compute Hessian $\begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}$ det	$\begin{pmatrix} 2 & -1 \\ -1 & -4 \end{pmatrix}$ det $2 \cdot (-4) - (-1)^2 = -9$ is < 0 <u>Saddle pt.</u>	$\begin{pmatrix} 6 & 1 \\ 1 & 2 \end{pmatrix}$ det is $2 \cdot 6 - 1^2 = 11$ is > 0 top left is 6 is > 0 <u>local minimum pt</u>	

$$f(x,y) =$$

c) $2x^4 + y^2 - xy + 1$

$$f_x(x,y) = 8x^3 - y$$

$$f_y(x,y) = 2y - x$$

At crit pts, get $\begin{cases} 8x^3 = y \\ 2y = x \end{cases}$

so $64y^3 = y$, i.e. $y = 0$ or $\pm \frac{1}{8}$

Corresp. $x = 0$ or $\pm \frac{1}{4}$

i.e. 3 crit pts $(x,y) = (\underline{0,0}), (\underline{\frac{1}{4}, \frac{1}{8}}), (\underline{-\frac{1}{4}, -\frac{1}{8}})$

Hessian matrix $\begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} 24x^2 & -1 \\ -1 & 2 \end{pmatrix}$

$\det(\text{Hessian})$ is $48x^2 - 1$

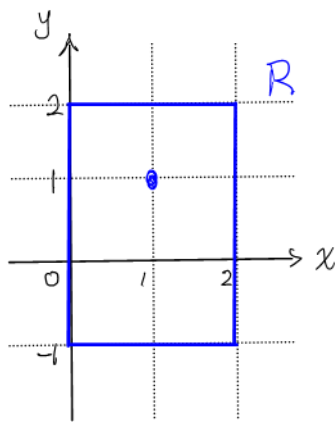
At $(0,0)$, $\det$ is -1 is < 0 , saddle pt.

At $(\frac{1}{4}, \frac{1}{8})$, $\det$ is $\frac{48}{16} - 1$ is ≥ 0 , btm right is 2 is > 0 , local minimum pt

At $(-\frac{1}{4}, -\frac{1}{8})$, $\det$ is $\frac{48}{16} - 1$ is ≥ 0 , btm right is 2 is > 0 , local minimum pt

a) Find the maximum and minimum points of the function $2x^2 - 2xy + y^2 - 2x$ on the rectangle $R: 0 \leq x \leq 2; -1 \leq y \leq 2$; using this information, determine the type of the critical point.

b) Confirm the critical point type by using the second derivative test.



$$f(x, y) := 2x^2 - 2xy + y^2 - 2x$$

$$f_x(x, y) = 4x - 2y - 2$$

$$f_y(x, y) = -2x + 2y$$

At critical pt of f , get $f_x = f_y = 0$
 i.e. $4x - 2y = 2$
 $-2x + 2y = 0$

i.e. solve, get
 $(x, y) = (1, 1)$

Hessian matrix $\begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} 4 & -2 \\ -2 & 2 \end{pmatrix}$

det is $8 - 4 = 4$ is > 0
 top left is 4 is > 0 } local minimum

only critical pt of f .

Have to restrict f to the four boundary line segments to find (non critical) max/min pts.

$$f(x, y) := 2x^2 - 2xy + y^2 - 2x$$

Now optimize these 4 bdy versions of f as 1 var fn

• $f = y^2$ for $y \in [-1, 2]$

Critical pt at $y = 0$, 2nd der. ≥ 0 , local min

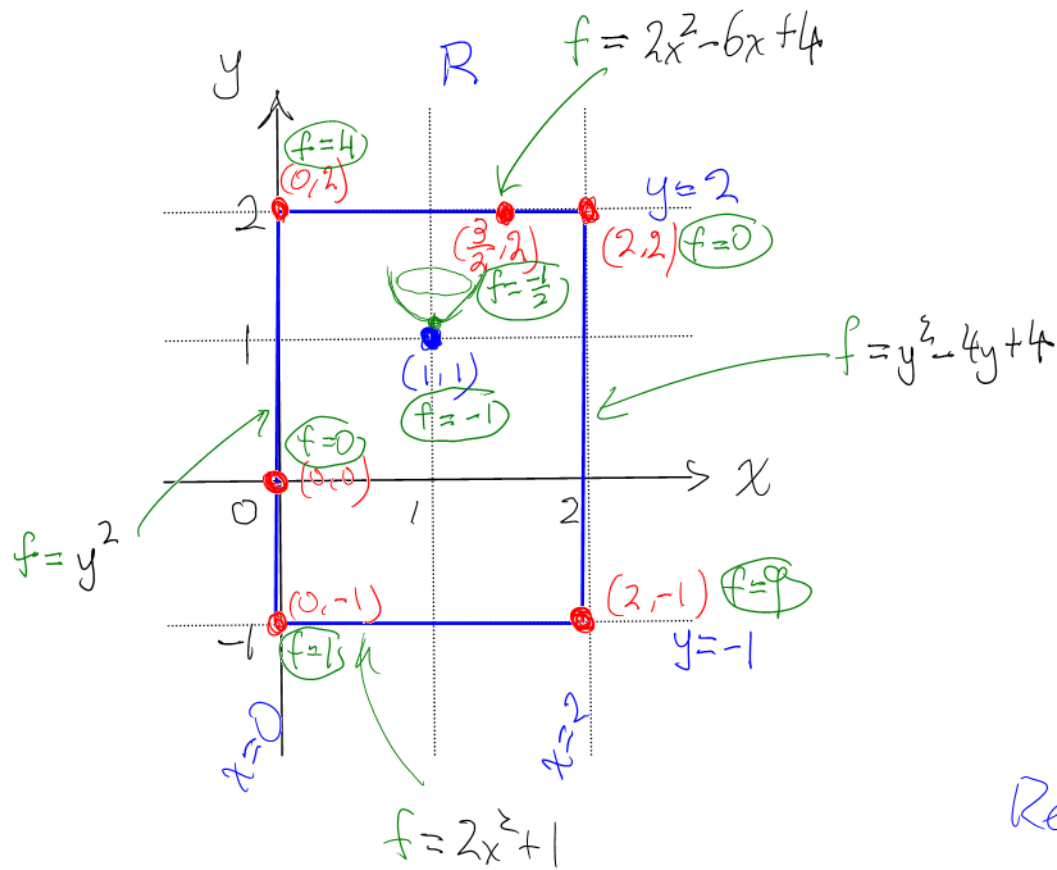
At bdy pts $y = -1$, fn f takes val. 2
 $y = 2$ -1

Repeat analysis for

• $f = y^2 - 4y + 4$ for $y \in [-1, 2]$

• $f = 2x^2 + 1$ for $x \in [0, 2]$

• $f = 2x^2 - 6x + 4$ for $x \in [0, 2]$



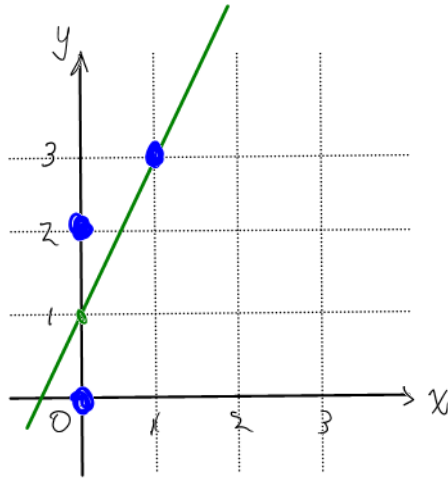
Global {max
min} of f on R

is $\begin{cases} f = 9 \\ f = -1 \end{cases}$ attained at $\begin{cases} (2, -1) \\ (1, 1) \end{cases}$

$$y = ax + b \quad \text{minimize} \quad D = \sum_{i=1}^3 (ax_i + b - y_i)^2$$

Find by the method of least squares the line which best fits the three data points

a) $(0,0), (0,2), (1,3)$



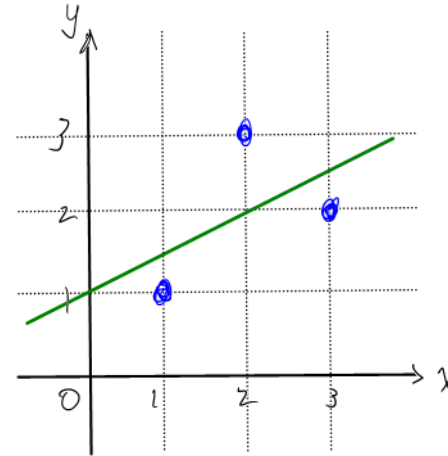
$$\begin{aligned} D &= (a \cdot 0 + b - 0)^2 + (a \cdot 0 + b - 2)^2 + (a \cdot 1 + b - 3)^2 \\ &= b^2 + (b-2)^2 + (a+b-3)^2 \\ &= a^2 + 2ab + 3b^2 - 6a - 10b + 13 \end{aligned}$$

$$\begin{aligned} D_a &= 2a + 2b - 6 \\ D_b &= 2a + 6b - 10 \end{aligned} \quad \left. \begin{array}{l} \text{At min for } D, \text{ get} \\ 2a + 2b = 6 \\ 2a + 6b = 10 \end{array} \right\}$$

Solve, get $a=2, b=1$

So best sq. line is $y = 2x + 1$

b) $(1,1), (2,3), (3,2)$



$$\begin{aligned} D &= (a \cdot 1 + b - 1)^2 + (a \cdot 2 + b - 3)^2 + (a \cdot 3 + b - 2)^2 \\ &= (a+b-1)^2 + (2a+b-3)^2 + (3a+b-2)^2 \end{aligned}$$

$$\begin{aligned} D_a &= 2(a+b-1) + 2(2a+b-3) \cdot 2 + 2(3a+b-2) \cdot 3 \\ &= 28a + 12b - 26 \end{aligned}$$

$$\begin{aligned} D_b &= 2(a+b-1) + 2(2a+b-3) + 2(3a+b-2) \\ &= 12a + 6b - 12 \end{aligned}$$

$$\text{At min for } D, \text{ get} \quad \left\{ \begin{array}{l} 28a + 12b = 26 \\ 12a + 6b = 12 \end{array} \right.$$

Solve, get $a = \frac{1}{2}, b = 1$

best sq line is $y = x/2 + 1$

$$y = f(x), \quad f \text{ deg 2 poly in } x$$

Use least squares to fit a second degree polynomial exactly through the points

$(-1, -1), (0, 0), (1, 3)$

Mthd 0 (not least square)

$$y = \underline{a}x^2 + \underline{b}x + \underline{c}$$

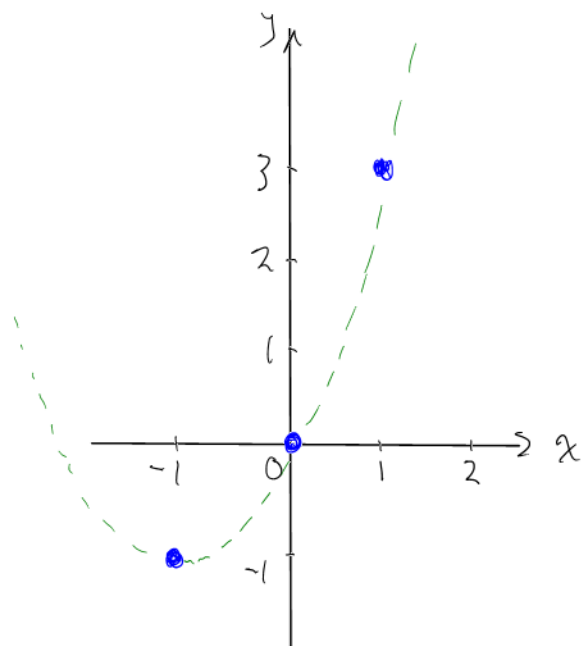
$$a, b, c \text{ must satisfy } -1 = a(-1)^2 + b(-1) + c$$

$$0 = a(0)^2 + b(0) + c$$

$$3 = a(+1)^2 + b(+1) + c$$

System of 3 ^(main) eqns in 3 unknowns a, b, c

solve using linear algebra $(a, b, c) = (1, 2, 0)$



Least Sq Mthd: minimize
(as fn of a, b, c)

$$\begin{aligned} D &= \left(a(-1)^2 + b(-1) + c - (-1) \right)^2 \leftarrow \text{sq of error at } x=-1 \\ &+ \left(a(0)^2 + b(0) + c - (0) \right)^2 \leftarrow \text{sq of error at } x=0 \\ &+ \left(a(+1)^2 + b(+1) + c - (3) \right)^2 \leftarrow \text{sq of error at } x=1 \\ &= (a-b+c+1)^2 + c^2 + (a+b+c-3)^2 \end{aligned}$$

Compute $\mathcal{D}_a = 2(a-b+c+1) + 2(a+b+c-3) = 4a + 4c - 4$

$$\mathcal{D}_b = -2(a-b+c+1) + 2(a+b+c-3) = 4b - 8$$

$$\mathcal{D}_c = 2(a-b+c+1) + 2c + 2(a+b+c-3) = 4a + 4b + 4c - 4$$

At min for $\mathcal{D}$, have $\mathcal{D}_a = \mathcal{D}_b = \mathcal{D}_c = 0$

$$\text{So } \left. \begin{array}{l} 4a + 4c = 4 \\ 4b = 8 \\ 4a + 4b + 4c = 4 \end{array} \right\} \text{ie. } b=2, c=0, a=1$$

ie. least sq poly is $x^2 + 2x = (x+1)^2 - 1$